



Surface Finish

Groover M.P.: Chapters 6 and 20



The student knows:

- what surface finish and integrity are;
- how to measure the surface finish (roughness and waviness);
- how to predict the surface finish due to a machining process.





WHY ARE SURFACES IMPORTANT?

3

- Aesthetic reasons
- Surfaces affect safety
- Friction and wear depend on surface characteristics
- Surfaces affect mechanical and physical properties
- Assembly of parts is affected by their surfaces
- Smooth surfaces make better electrical contacts

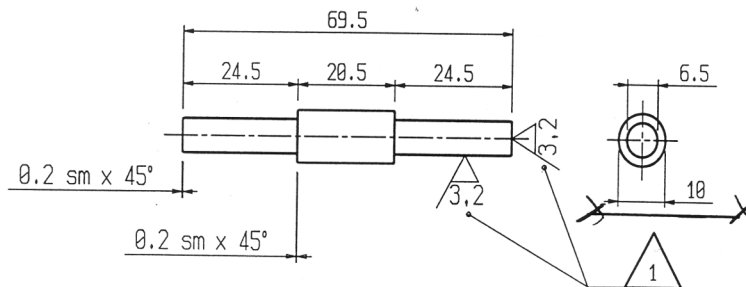


Cloud gate (A. Kapoor, 2006)
Millennium Park - Chicago
Stainless Steel



Ideal Surface: Designer's intended surface contour of part, defined by lines in the engineering drawing

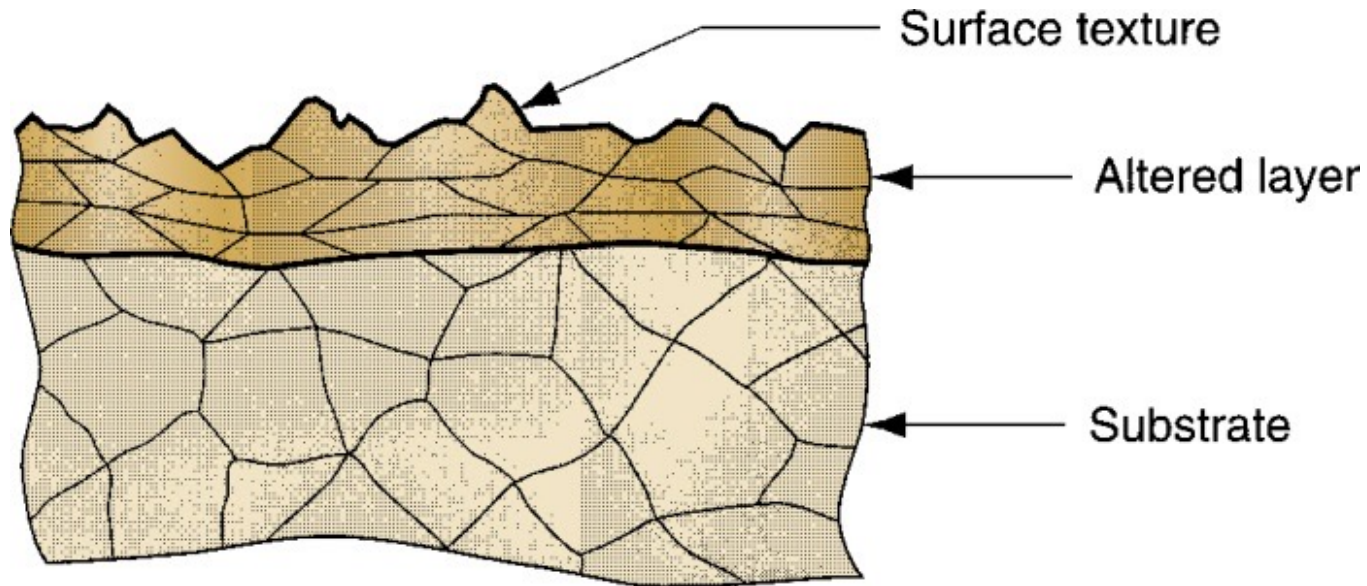
- Ideal surfaces appear as absolutely straight lines, ideal circles, round holes, and other edges and surfaces that are geometrically perfect
- Actual surfaces of a part are determined by the manufacturing processes used to make them
- Variety of processes result in wide variations in surface characteristics



"90034" design Christopher Dresser, 1885



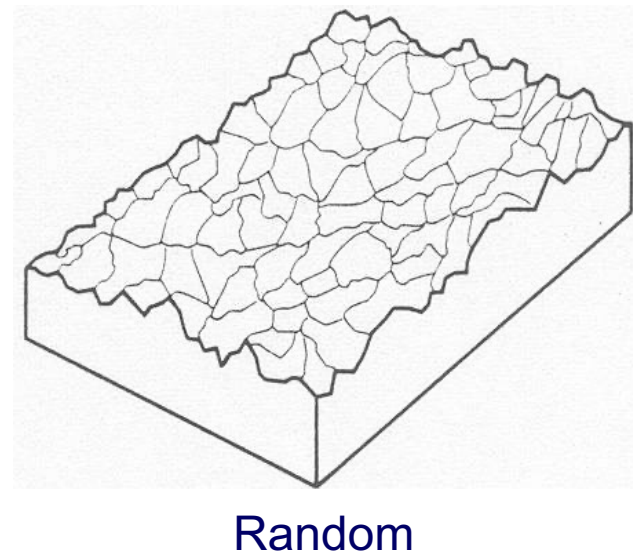
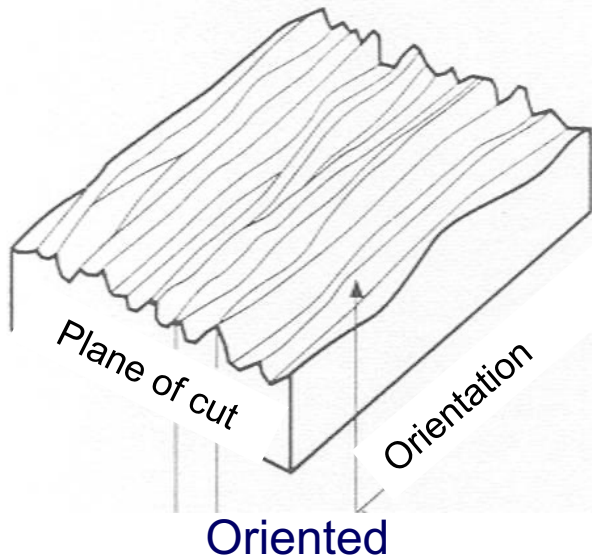
Magnified cross section of a typical metallic part surface





The topography and geometric features of the surface

- When highly magnified, the surface is anything but straight and smooth. It has roughness, waviness, and flaws.
- It also possesses a pattern and/or direction resulting from the mechanical process that produced it.

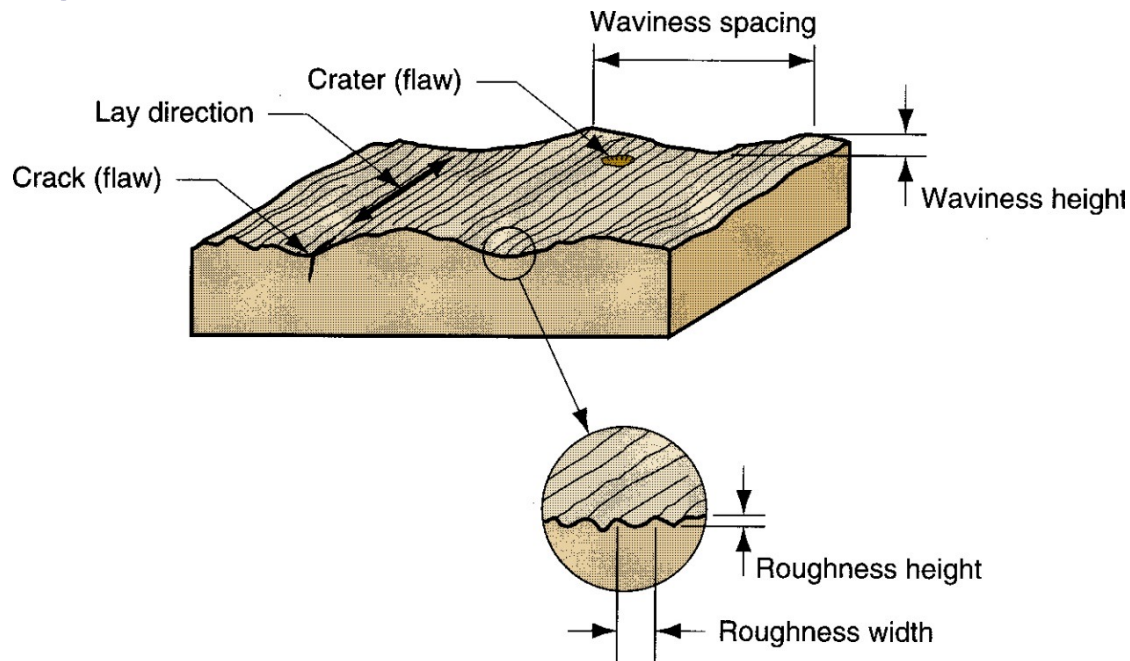




FOUR ELEMENTS OF SURFACE TEXTURE

7

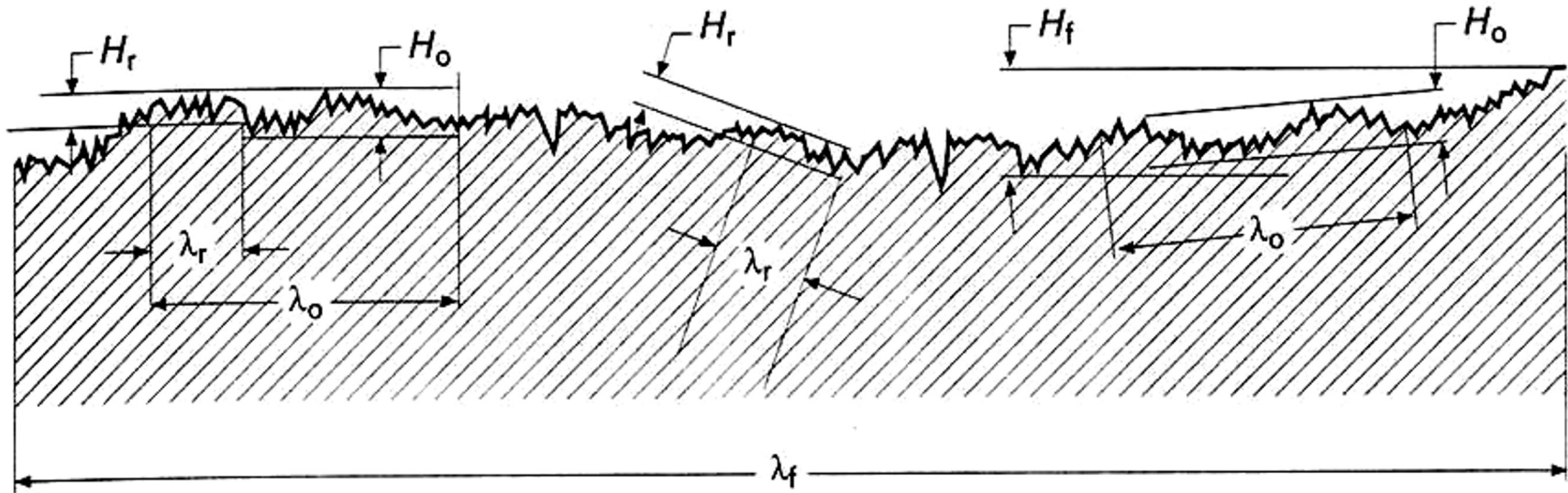
1. Roughness: small, finely-spaced deviations from ideal surface
 - Determined by material characteristics and processes that formed the surface.
2. Waviness: deviations of much larger spacing
 - Waviness deviations occur due to work deflection, vibration, tooling, and similar factors,
 - Roughness is superimposed on waviness.





FOUR ELEMENTS OF SURFACE TEXTURE

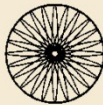
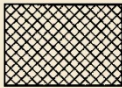
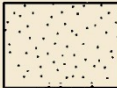
8



Type of deviation		Ratio wavelength/height
Microgeometric deviations	Roughness or primary texture	$0 < \lambda_r/h_r \leq 50$
	Waviness or secondary texture	$50 < \lambda_o/h_o \leq 1000$
Macrogeometric deviations	Form errors	$1000 < \lambda_f/h_f$



3. Lay - predominant direction or pattern of the surface texture

Lay symbol	Surface pattern	Description	Lay symbol	Surface pattern	Description
=		Lay is parallel to line representing surface to which symbol is applied.	C		Lay is circular relative to center of surface to which symbol is applied.
⊥		Lay is perpendicular to line representing surface to which symbol is applied.	R		Lay is approximately radial relative to the center of the surface to which symbol is applied.
X		Lay is angular in both directions to line representing surface to which symbol is applied.	P		Lay is particulate, nondirectional, or protuberant.

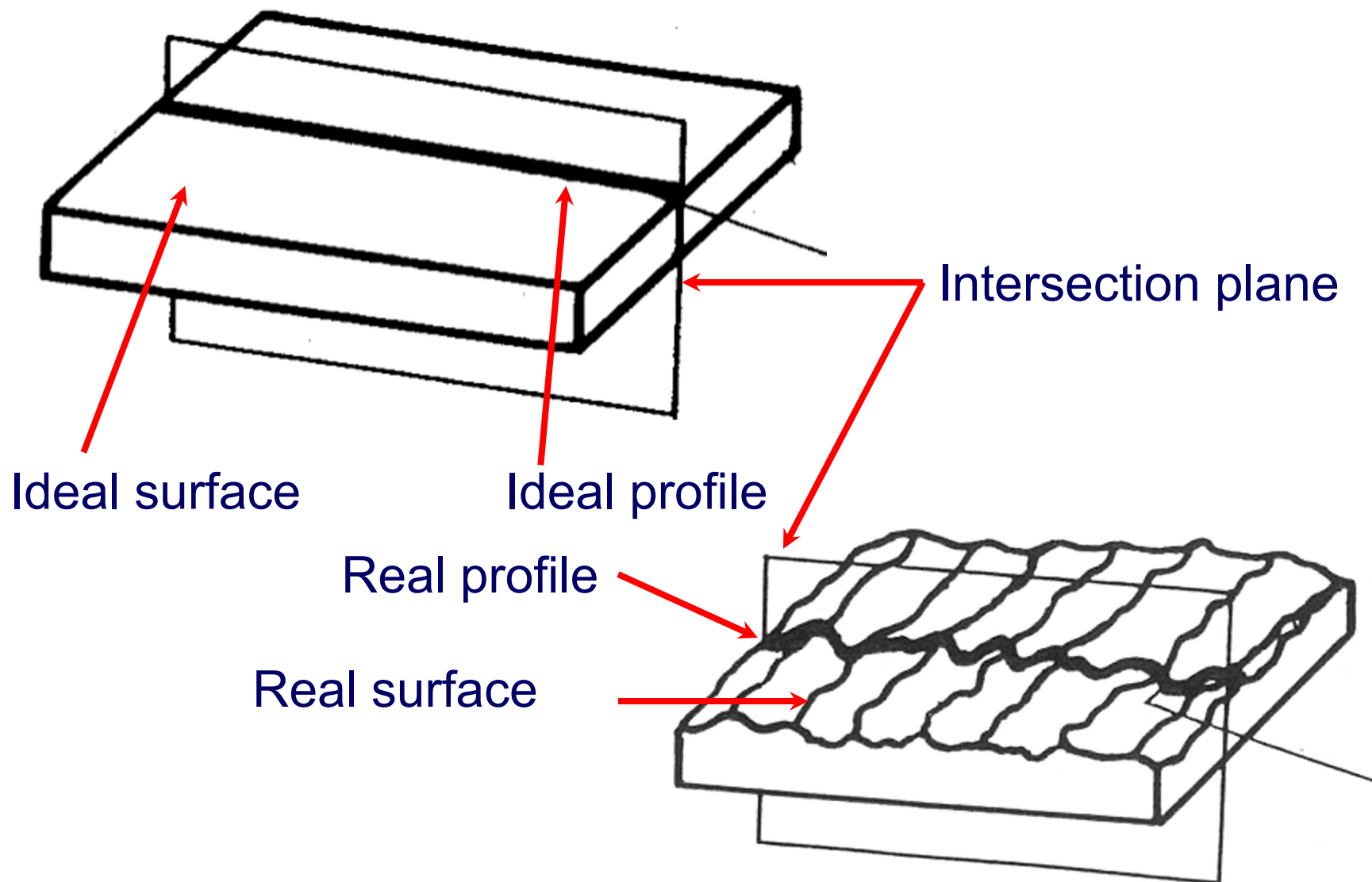
4. Flaws - irregularities that occur occasionally on the surface

- Includes cracks, scratches, inclusions, and similar defects in the surface
- Although some flaws relate to surface texture, they also affect surface integrity



The following surfaces must be considered in the study of the surface texture:

- **Ideal surface:** surface as designed by the designer
- **Real surface:** surface limiting the body and separating it from the surrounding medium (it is generated by the manufacturing process)
- **Primary Surface:** Surface probed by means of a measuring system
- **Reference surface:** Surface with reference to which the real surface is measured
- **Mean surface:** Surface identical to the reference surface, but located as resulting from a least squares fitting





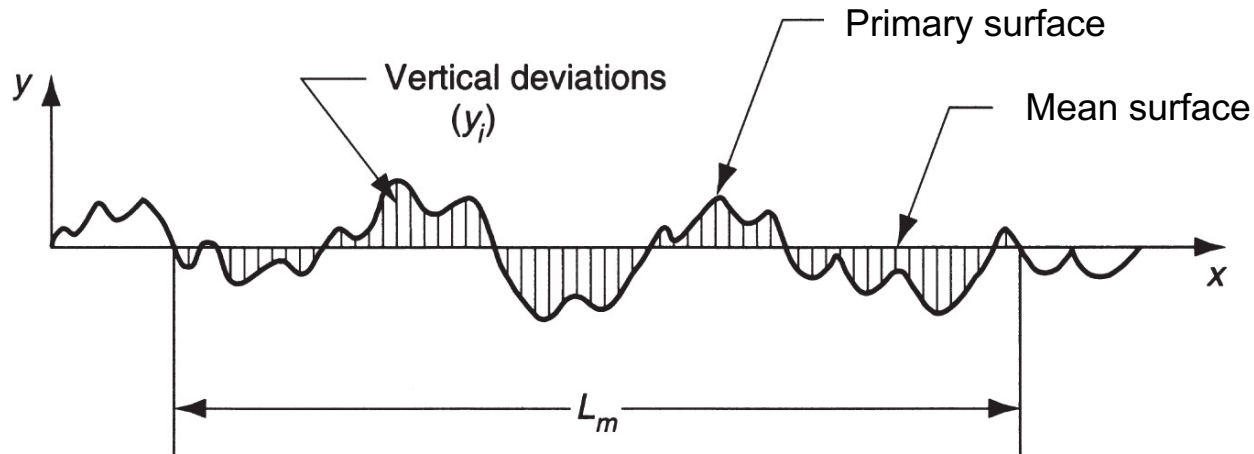
Surface roughness: a measurable characteristic based on roughness deviations

Surface finish: a more subjective term denoting smoothness and general quality of a surface

- In popular usage, surface finish is often used as a synonym for surface roughness
- Both terms are within the scope of surface texture



Vertical deviations from mean surface over a specified surface length

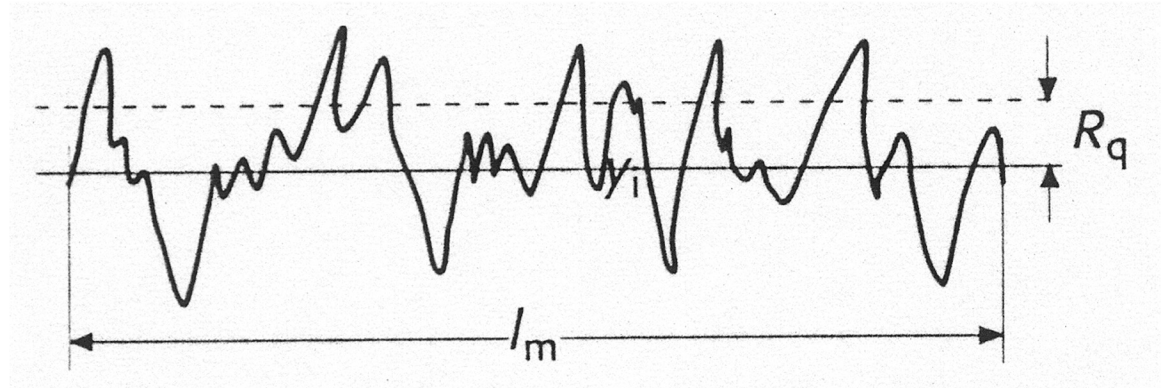


Arithmetic average based on absolute values of deviations, and it is referred to as *average roughness* (R_a)

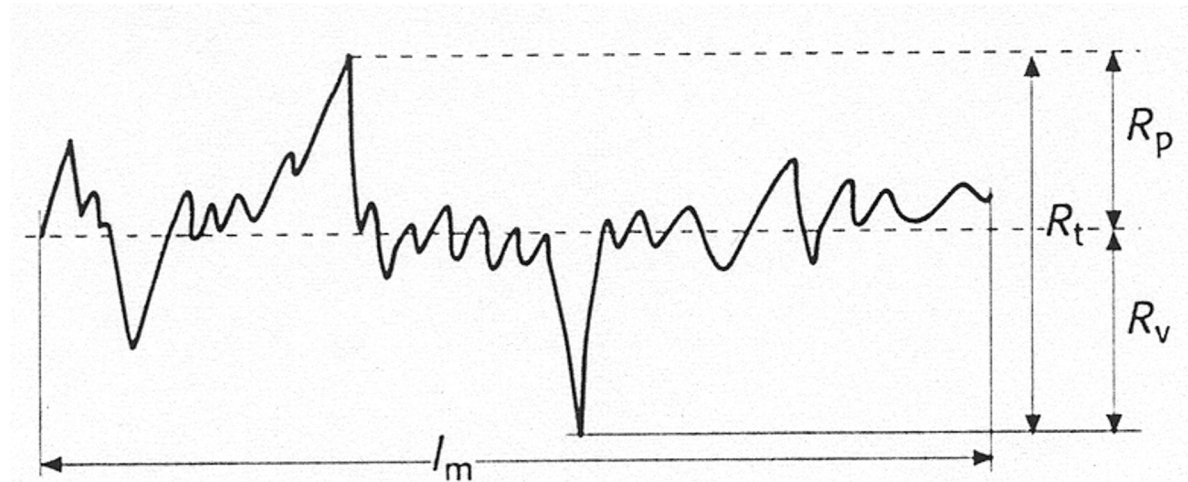
$$R_a = \int_0^{L_m} \frac{|y|}{L_m} dx \quad R_a = \sum_{i=1}^n \frac{|y_i|}{n}$$



$$R_q = \sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2}$$

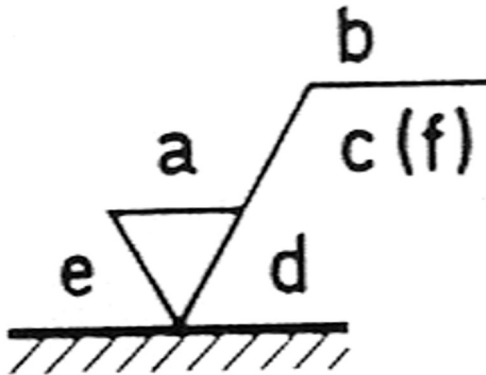


$$R_t = R_p + R_v$$





Surface texture symbols in engineering drawings

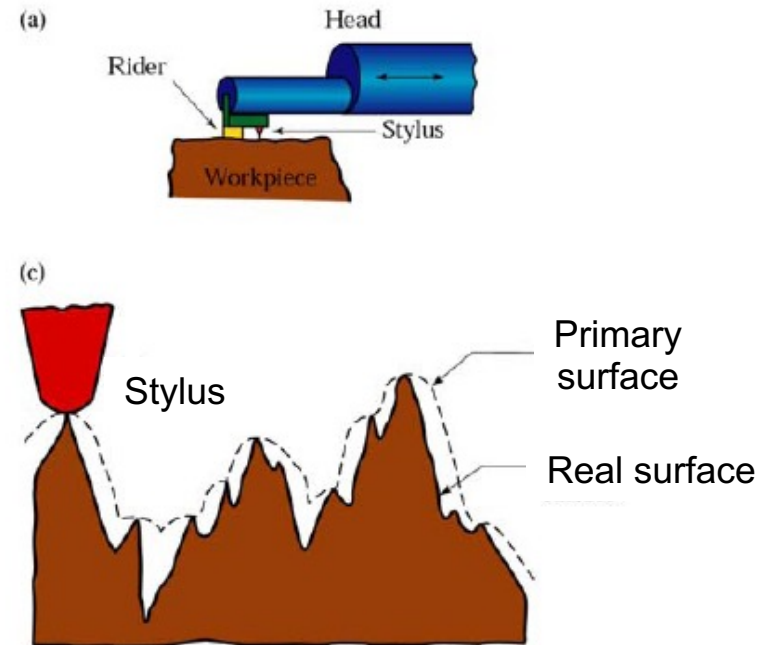
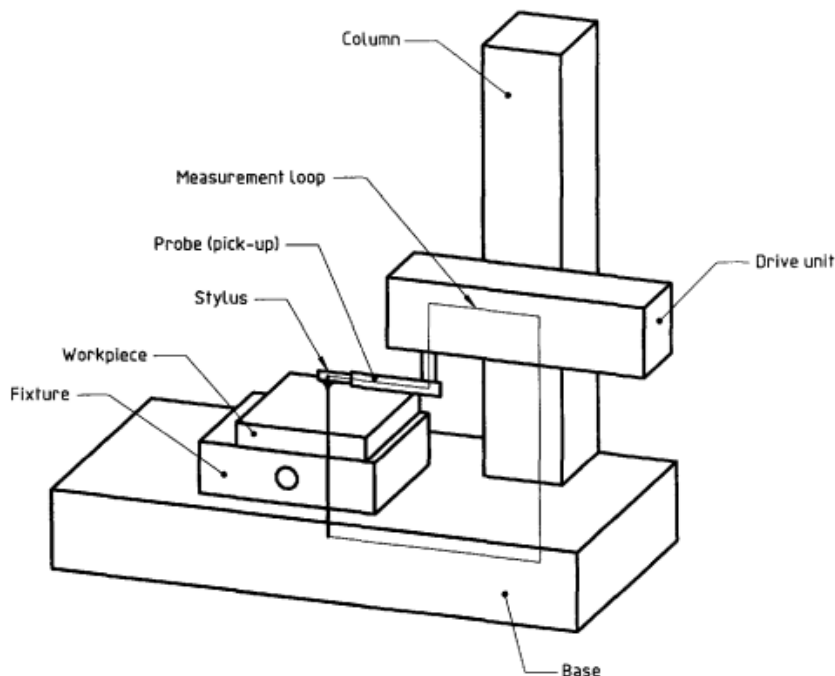


- a) R_a or roughness grade;
- b) waviness;
- c) cutoff length;
- d) lay symbol;
- e) machining allowance;
- f) other roughness parameters.



Three methods to measure surface roughness:

1. Subjective comparison with standard test surfaces
 - Fingernail test
2. **Stylus electronic instruments**
3. Optical techniques

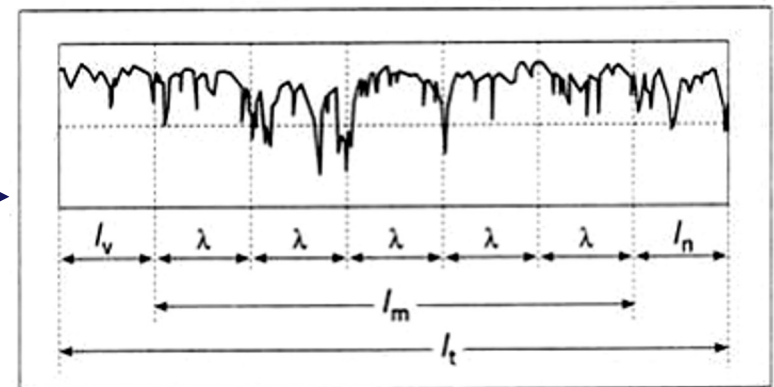
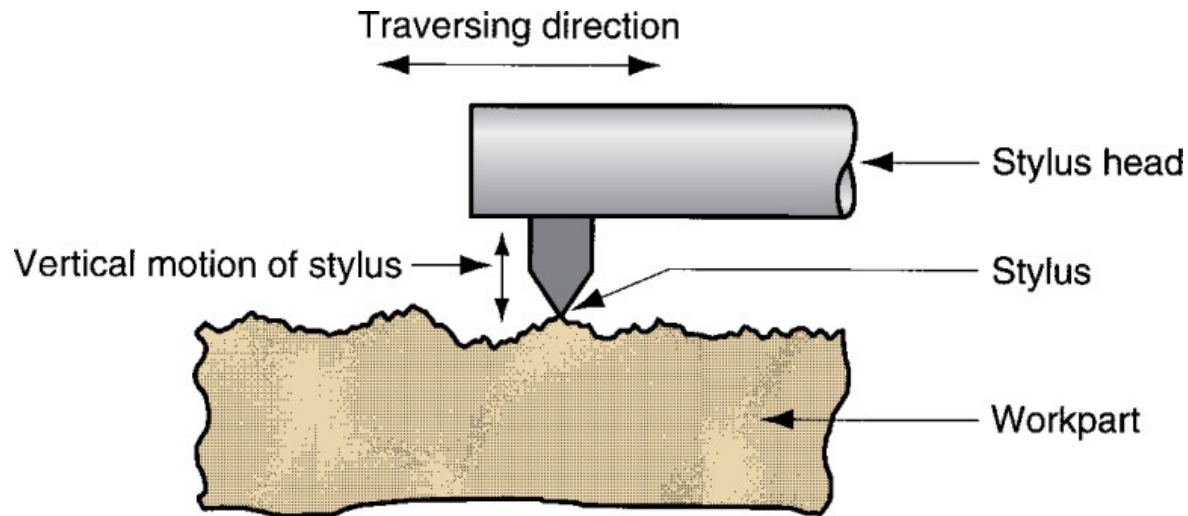




MEASUREMENT OF SURFACE ROUGHNESS

17

Stylus head traverses horizontally across surface, while stylus moves vertically to follow surface profile.

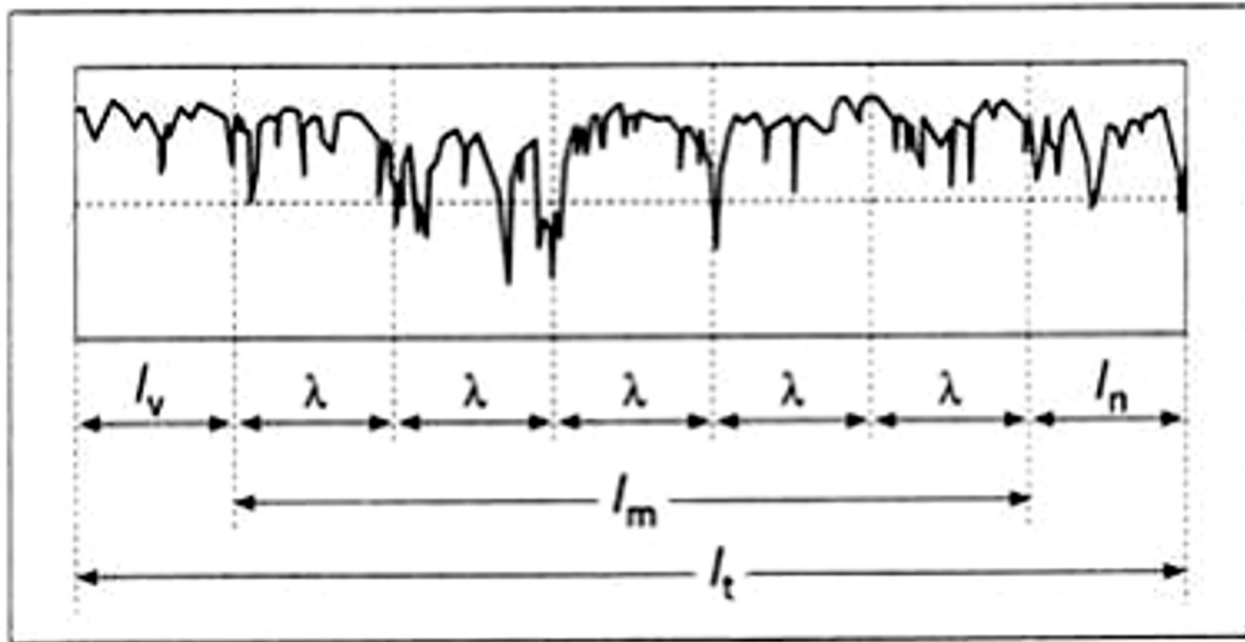




A problem with the R_a computation is that waviness may get included. To deal with this problem, a parameter called the **cutoff length** is used as a filter to separate waviness from roughness deviations.

Cutoff length is a sampling distance along the surface:

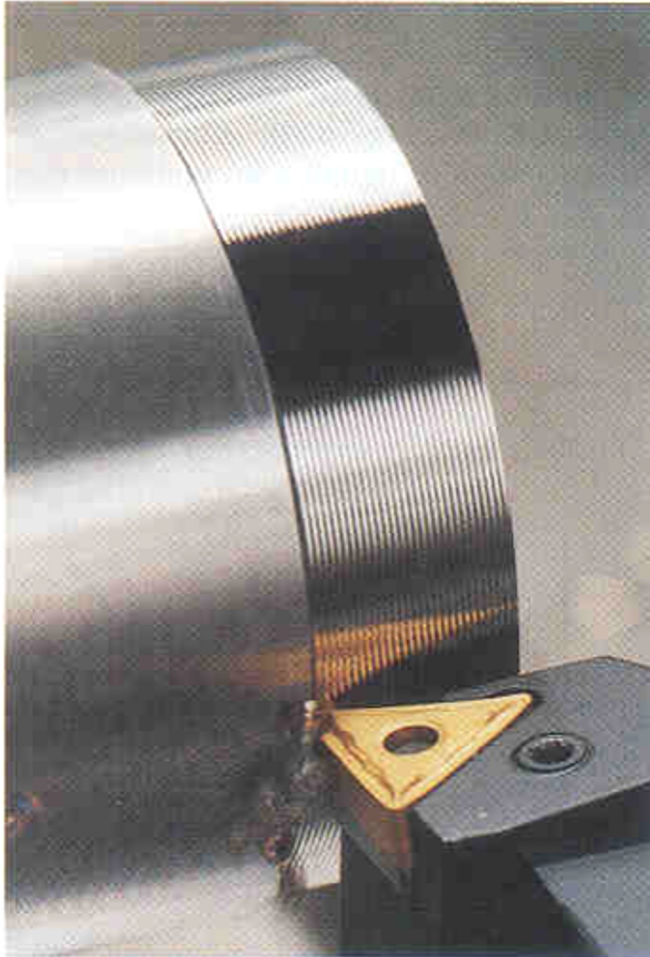
- A sampling distance shorter than the waviness eliminates waviness deviations and only includes roughness deviations.





SURFACE ROUGHNESS IN TURNING

19





SURFACE ROUGHNESS IN TURNING

- **Theoretical roughness:** roughness that can be computed theoretically starting from the geometrical characteristics of the process.
- **Real roughness:** actual roughness of the machined part.



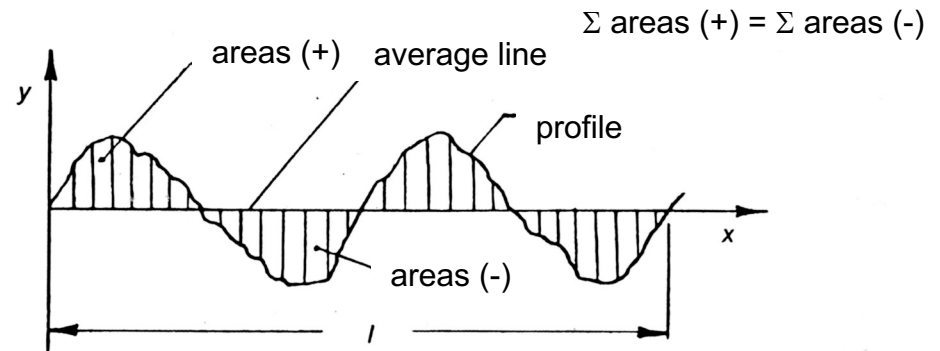
SURFACE ROUGHNESS IN TURNING

Roughness Parameters

- R_{max} or R_t : “maximum distance (expressed in μm) between the highest peaks and the deepest valleys”.
- R_a : “arithmetic average (expressed in μm) of the absolute values of the deviations y of the real profile from the average line”.

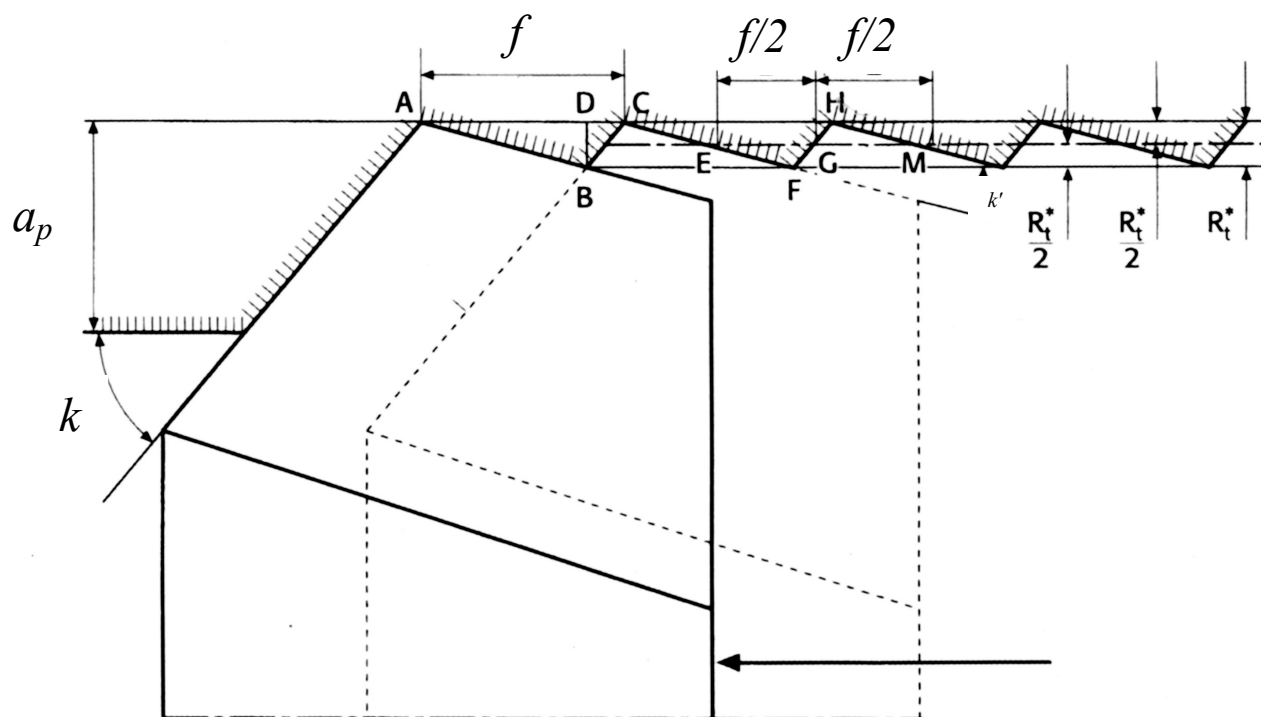
$$R_a = \frac{1}{l} \int_0^l |y| \cdot dx$$

$$R_a \cong \frac{R_{max}}{4}$$



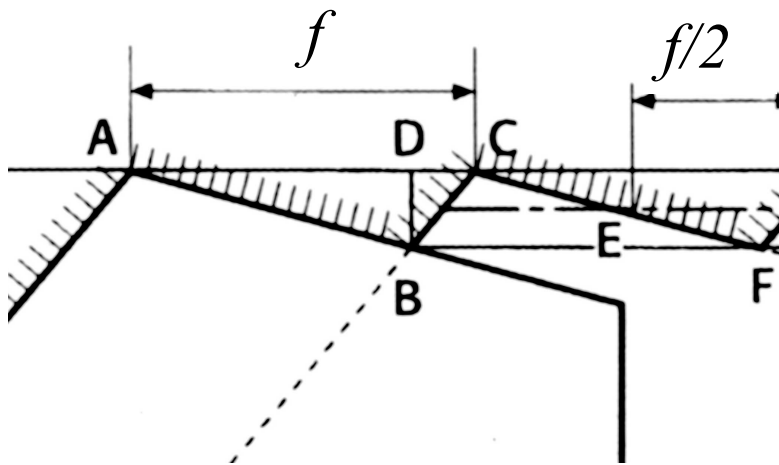


Theoretical roughness: tool with zero nose radius





Theoretical roughness: tool with zero nose radius



f is given by:

$$f = \overline{AD} + \overline{DC}$$

$$f = \overline{DB} (\cotg k' + \cotg k)$$

since:

$$\overline{DB} = R_t = R_{\max}$$

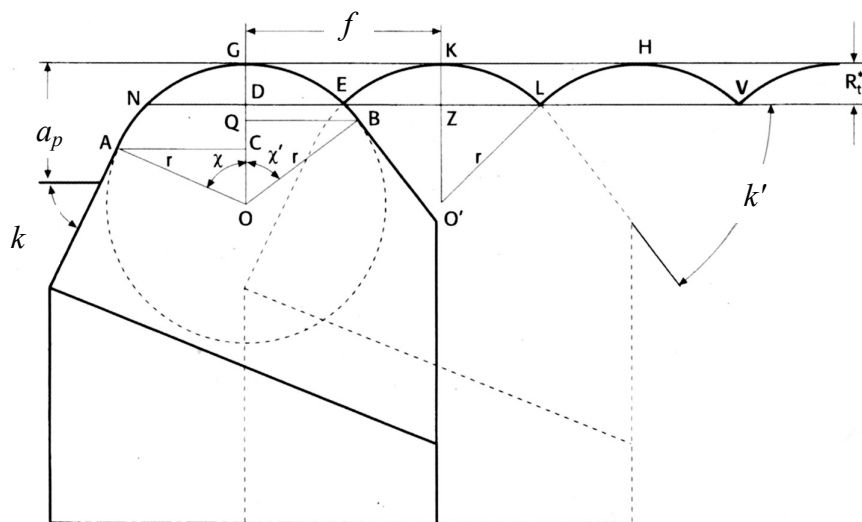
It follows that:

$$R_{\max} = \frac{f}{\cotg k' + \cotg k} \cdot 10^3 [\mu\text{m}] \quad R_a = \frac{R_{\max}}{4}$$



SURFACE ROUGHNESS IN TURNING

Theoretical roughness: tool with nose radius > 0



since:

$$\overline{ND} \leq \overline{AC}$$

$$\overline{DE} \leq \overline{QB}$$

$$\overline{ND} = \overline{DE} = \frac{f}{2}$$

$$\frac{f}{2} < r \cdot \sin(k') \Rightarrow f < 2r \cdot \sin(k')$$

$$\Rightarrow \frac{f}{2} < r \cdot \sin(k) \Rightarrow f < 2r \cdot \sin(k)$$

Then we obtain:

Approx. **Schmaltz** formula

$$R_{\max} = R_t = \overline{OG} - \overline{OD} = \overline{OG} - \sqrt{\overline{ON}^2 - \overline{ND}^2} = \left(r - \sqrt{r^2 - \frac{f^2}{4}} \right) 10^3 \approx \frac{f^2}{8 \cdot r} 10^3 \quad [\mu\text{m}]$$



SURFACE ROUGHNESS IN TURNING

Effects of feed f and nose radius r

$$R_a = \frac{1000}{32} \frac{f^2}{r}$$

